Experiment – 06 (HomeTask)

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Semester: 5th
Subject: Advanced Database Management System

Question 1: Placements

Source: HackerRank — Placements — https://www.hackerrank.com/challenges/placements/problem

You are given three tables: Students, Friends, and Packages. Students contains the ID and name of each student. Friends contains the ID of each student and the ID of their best friend. Packages contains the ID of each student and the amount of the package offered to them (in lakhs). Write a query to output the names of those students whose best friends got offered a higher salary package than them, ordered by the salary amount offered to their best friends.

Solution:

```
SELECT S.NAME
FROM STUDENTS AS S
JOIN FRIENDS AS F
  ON S.ID = F.ID
JOIN PACKAGES AS P
  ON S.ID = P.ID
JOIN PACKAGES AS FP
  ON F.FRIEND_ID = FP.ID
WHERE FP.SALARY > P.SALARY
ORDER BY FP.SALARY;
```

Question 2: Symmetric Pairs

Source: HackerRank — Symmetric Pairs — https://www.hackerrank.com/challenges/symmetric-pairs/problem

You are given a table, Functions, containing two columns: X and Y. Two pairs (X1, Y1) and (X2, Y2) are said to be symmetric pairs if X1 = Y2 and X2 = Y1. Write a query to output all such symmetric pairs in ascending order by the value of X. List the identical pairs (where X = Y) only once.

Solution:

```
SELECT
  F1.X,
  F1.Y
FROM FUNCTIONS AS F1
JOIN FUNCTIONS AS F2
  ON F1.X = F2.Y AND F1.Y = F2.X
WHERE F1.X <= F1.Y
GROUP BY F1.X, F1.Y
HAVING COUNT(*) > 1 OR F1.X = F1.Y
ORDER BY F1.X;
```